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ORIGINAL PAPER

Development of a paediatric population pharmacokinetic model for valacyclovir from literature non-compartmental values originating from sparse studies and Bayesian priors: a simulation study

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Abstract A preliminary population pharmacokinetic (PopPK) model of valacyclovir in children was developed from non-compartmental analysis (NCA) parameter values from literature, including several age groups, combined with Bayesian priors from a PopPK model of acyclovir, the active metabolite of valacyclovir, from literature too. Also a simulation study was carried out to evaluate the performance of various modelling choices related to the estimation of model parameters from NCA parameters originating from sparse PK studies. Assuming a one-compartment model with first order absorption, a mixed effects, meta-analysis approach was utilized which allows accounting the random intergroup variability, the detection of covariates and the application of informative Bayesian priors on the parameters. The conclusions from the simulation study calculating bias and precision for various cases, were that a model which takes explicitly into account the sampling schedule, performs better than a model using the theoretical expressions of calculating the NCA parameters. Also by using the geometric rather than the arithmetic means of NCA parameters, less biased results are obtained. These findings guided the choices for the valacyclovir model, for which informative priors from a PopPK model of acyclovir were applied for some of the parameters, in order to include a richer covariate model for clearance, not supported by the NCA dataset and a value for bioavailability. This preliminary valacyclovir model can be used in simulations to provide dosage recommendations for children of various ages and to help design more efficiently prospective clinical trials.

Keywords Bayesian priors · Meta-analysis · Paediatric pharmacokinetic model

Introduction

Modelling and simulation (M&S) is upgrading its role in drug development and has started to be encouraged by regulators, in order to better exploit existing data and design more efficiently prospective clinical studies, for sensitive populations, such as children where there are more strict ethical limitations for clinical trials. In this context EMA is encouraging the usage of M&S to support the development of paediatric formulations [1]. One of the main tools to perform M&S studies is by carrying out Monte Carlo simulations with population pharmacokinetic (PopPK) models. One limitation to carry out simulation studies is the availability of PopPK models for the drug, formulation and population of interest, in literature. Although, hundreds of publications have appeared in the recent years describing PopPK models for a vast range of drugs, still there are many cases for which no PopPK model is available in literature. In such cases, a solution is to attempt to compile a PopPK model from literature data, with some sort of meta-analysis methodology. In pharmacokinetic studies available in literature, very rarely can raw data from individual patients be found, even in plots that can be digitized. In most cases what is available, are tables of non-compartmental analysis (NCA) parameters. In the past, a few papers have appeared, that address this problem.

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In 2005, Dansirikul et al. [2] presented a methodology to transform NCA parameters to compartmental, based on Microsoft Excel, and in 2010, Wang et al. [3] presented a meta-analysis mixed effects based methodology for the same task. Both methods work reasonably well, as the authors demonstrate, mostly for NCA parameter values coming from dense datasets. PK studies with sparse sampling such as paediatric studies, are more difficult to analyze. One of the potential problems is that the reported NCA parameter values may include bias, which will propagate to the estimated compartmental parameter values.

Valacyclovir is a prodrug of acyclovir an antivirial drug indicated for herpes simplex (HSV) and varicella zoster (VZV) viruses. While acyclovir is approved for paediatric patients, valacyclovir is not, at least in Europe. In order to carry out a simulation study that will help to determine the appropriate doses of oral valacyclovir that are therapeutically equivalent to the approved doses of acyclovir in paediatric patients, and also design a prospective clinical trial aiming to produce new clinical PK data in paediatric patients that will allow the development of a new PopPK model from raw data, a preliminary PopPK model for oral valacyclovir in children was needed. The only available PopPK model for valacyclovir in literature was the work of Zeng et al. [4]. However, this model is for cancer paediatric patients and its results seem to diverge considerably from the PK parameters of valacyclovir found in literature, as the authors of this work admit, possibly because of the characteristics of this special population. As a result [4] was considered inadequate to be used for a general paediatric population. A PopPK model for acyclovir is also available, found in the work of Tod et al. [5]. This latter model is useful as valacyclovir is a prodrug of acyclovir, but still the process of appearance of acyclovir in the body after oral valacyclovir administration is absent. The solution chosen, to fill the gap of a PopPK model for valacyclovir, was to develop a model from literature data, namely from published NCA parameter values, published in accordance with Article 45 of the pediatric regulation of EMA [6] and further discussed by Kimberlin et al. in [7] and integrate this information with the findings of [5] using a Bayesian approach.

The present work is a result of the pursuit to develop a usable PopPK model for acyclovir after per os valacyclovir administration, in children from NCA parameter values found in literature. It is separated in two parts: (i) A simulation study to evaluate the performance of the various modelling choices that had to be made to develop the valacyclovir model from NCA values coming from PK studies with sparse sampling, such as different structural models, various choices related to the data, the effect of certain assumptions made and other questions. The simulations were designed to mimic the available datasets of valacyclovir, including the sparse sampling schedule of the original PK study. Among the options that were tested, were two different structural models, a model based on calculating the NCA parameter values from the theoretical formulas similarly to [2, 3] and a second model based on calculating the NCA parameter values taking into account the sampling times. This latter technique was used to cancel out the potential bias introduced by the sparse sampling schedule and has been used before in [2] for a two compartment model with dense sampling. The methodology was implemented using a mixed effects approach. (ii) The second part of the paper is the actual application of the findings of the simulation study to the valacyclovir dataset [6] in order to develop a PopPK model. As it is demonstrated in the results one, of the advantages of the mixed effects methodology presented here is the potential to apply informative Bayesian priors on the parameters in order to integrate to the model, additional information available in the form of parameter values. In the valacyclovir model, informative priors from the PopPK model of acyclovir from the paper of Tod et al. [5] were applied.

Methods

We assume that means and/or geometric means of NCA parameters are available for the kinetics of a drug, possibly taken from literature, namely the maximum concentration, C_{max}, the area under the concentration curve (AUC), the time when the maximum concentration was observed, T_{max} and the half-life, $t_{1/2}$. Several sets of values corresponding to different groups of patients, may be available, potentially with different characteristics, such as different age groups. The PK of the drug is assumed to follow a one-compartment model with first order absorption and elimination, parameterized as clearance (CL), volume of distribution (V) and absorption rate constant (k_a) with potentially a lag time (t_{lag}) for the initiation of absorption.

A two stage approach is applied to model the means and coefficients of variation of the non-compartmental parameters. Initially, the means are used to estimate the mean PopPK parameters. Two different structural models that calculate the NCA parameters are considered. In the second stage, inter-individual variability of the PopPK parameters are estimated from the coefficients of variation of the non-compartmental parameters conditional to the means of the first stage.

Structural models for the means

Two different structural models are considered to calculate the NCA parameters, namely C_{max}, AUC, T_{max} and $t_{1/2}$.

Model A where it is assumed that these are given by their theoretical values as calculated by the one-compartment model and model B where a known sparse sampling schedule is considered.

Model A: theoretical NCA values

For the one-compartment model the concentration of the drug at any time, t, is given by the following equation

$$\mathrm{C} = \frac{\mathrm{D}\cdot \mathrm{k_a}}{\mathrm{V}\left(\mathrm{k_a} - \frac{\mathrm{CL}}{\mathrm{V}}\right)}\left(\exp\left(-\frac{\mathrm{CL}}{\mathrm{V}}\cdot(\mathrm{t}-\mathrm{t_{lag}})\right) - \exp\left(-\mathrm{k_a}\cdot(\mathrm{t}-\mathrm{t_{lag}})\right)\right), \quad \text{for } t > \mathrm{t_{lag}} \tag{1}$$

Then the four NCA parameters of interest are given, respectively, by the following equations:

$$\mathrm{T_{max}} = \left(\log \mathrm{k_a} - \log\frac{\mathrm{CL}}{\mathrm{V}}\right) \Big/ \left(\mathrm{k_a} - \frac{\mathrm{CL}}{\mathrm{V}}\right) + \mathrm{t_{lag}} \tag{2}$$

$$\mathrm{C_{max}} = \frac{\mathrm{D}\cdot \mathrm{k_a}}{\mathrm{V}\left(\mathrm{k_a} - \frac{\mathrm{CL}}{\mathrm{V}}\right)}\cdot\left(\exp\left(-\frac{\mathrm{CL}}{\mathrm{V}}\cdot(\mathrm{T_{max}}-\mathrm{t_{lag}})\right) - \exp\left(-\mathrm{k_a}\cdot(\mathrm{T_{max}}-\mathrm{t_{lag}})\right)\right) \tag{3}$$

$$\mathrm{AUC} = \mathrm{D}/\mathrm{CL} \tag{4}$$

$$\mathrm{t_{1/2}} = \log 2\cdot \mathrm{V}/\mathrm{CL}. \tag{5}$$

Model B: calculated NCA values taking into account sampling times

Model B is constructed to calculate the NCA parameters C_{max}, AUC and $t_{1/2}$ exactly as these were derived from the raw data. Taking into account the sampling scheme, concentrations y_j are simulated at time points t_j from Eq. 1. The NCA parameters are then calculated from the simulated concentrations y_j as follows: $C_{max} = \max(y_i)$, the maximum of the y_j, $t_{1/2} = -ln2/slope$, where the slope is obtained by linear regression of the log transformed concentrations for at least the three last time points, AUC_{0-t_n} is calculated by the trapezoidal rule for the ascending part of the curve and by the log trapezoidal for the descending part, using the points (t_j, y_j), and extrapolating to infinity using the slope as defined previously, i.e.

$$\mathrm{AUC} = \mathrm{AUC_{0-t_n}} + \mathrm{C_n}/(-\mathrm{slope}). \tag{6}$$

For T_{max} the theoretical Eq. 2 is used for model B, too, since the real T_{max} according to the sampling scheme is a discrete variable which is more difficult to be combined with the others. Ignoring T_{max} was also examined as described below, in the simulations section. The two models may give similar predictions when the sampling scheme is dense enough but in a sparse sampling scheme there are notable differences.

Further, since different groups of patients may be available, random intergroup variability (ETAs) for the pharmacokinetic parameters, CL, V and k_a, using the exponential model, is considered, which can be used to identify important covariates, such as age or weight. Residual variability is modeled by the exponential error.

Model for the coefficients of variation

At a second stage, the inter-individual variability of the PK parameters, CV_{CL}, CV_V and CV_{k_a}, is estimated. A Fortran subroutine was built that predicts both the means and the coefficients of variation of the NCA parameters as a function of CV_{CL}, CV_V and CV_{k_a}. This function uses also the posthocs per group, of the mean values of CL, V and k_a, estimated in the first stage. The model implemented in the Fortran subroutine is fitted with NONMEM to both the means and the CVs of the NCA parameters in order to estimate the IIV of the PK parameters,CV_{CL}, CV_V and CV_{k_a}.

In the Fortran subroutine, two models were implemented, one for the means and one for the CVs of the NCA parameters. For the means of the NCA parameters, the structural model from the first stage is used, either model A or B. For the CVs of the NCA parameters, lognormally distributed inter-individual variability on the pharmacokinetic parameters CL, V and k_a is considered and the propagation of this IIV on the NCA parameters was implemented by Monte Carlo simulation. The empirical Bayes estimates (posthocs) per group, obtained from the first stage are used as mean PK parameters when generating the distributions of the PK parameters, while the variances of the distributions are calculated from CV_{CL}, CV_V and CV_{k_a} that are to be estimated.

The IIV distributions of the PK parameters are simulated as follows: The pharmacokinetic parameters CL, V and k_a are considered to be independently, lognormally distributed as $p_{ki} = \theta_k e^{\eta_i}$, where p_{ki} is the vector of the PK parameters for individual i of the group k, θ_k is the vector of the population means for group k and η_i follows the normal distribution with mean equal to zero and variance Ω. The mean of each distribution, θ_k is given by the empirical Bayes estimates (posthocs) obtained from the first stage while the variance of the distribution, Ω, is calculated from the parameters to be estimated, i.e. CV_{CL}, CV_V and CV_{k_a} and is assumed constant across the groups. Fifty thousand Monte Carlo samples of the distributions of the PK parameters are generated for each group. The NCA parameters for each simulated individual i are then calculated as in the first stage (by either model A or B) and the coefficients of variation are obtained. If a covariate for intergroup variability has been detected in the first stage, this

covariate is considered to apply also within each group as interindividual variability and is taken into account in the simulation. Two separate exponential residual error terms are assumed, one for the means and one for the coefficients of variation.

Estimation in both stages was performed in NONMEM V7.2.0 compiled with gfortran. FOCE method was used in the first stage and both models A and B were written in $PRED (supplementary information for NONMEM code). In the second stage, a custom Fortran subroutine was used to model the coefficients of variation of the NCA parameters as described above (supplementary information for Fortran code). Note that the random numbers of the Monte Carlo simulation were the same ones at each iteration to avoid discontinuities in the objective function that could cause problems in gradient based optimization method.

Simulation study

Simulated data

To assess the performance of the two structural models, A and B, and the overall limitations of the methodology, a simulation study was carried out. The scenario considered was that NCA parameters, namely C_{max}, T_{max}, AUC and $t_{1/2}$ are reported for a group of individuals as geometric (or arithmetic) means and CVs. Multiple groups provide a set of means and CVs for the NCA parameters which can be used to estimate the compartmental parameters. We assumed that the dataset consisted of ten groups of patients, and each group included ten individuals. Hundred such NCA parameter datasets were generated, from which the accuracy and precision for estimating the compartmental pharmacokinetic parameters were calculated.

In total three different dataset scenarios were considered, namely dataset I with 5 % residual error without t_{lag}, dataset II with 10 % residual error without t_{lag} and dataset III with 5 % residual error with t_{lag} (Table 1). To generate each dataset the following process was applied. Assuming one-compartment model with first order absorption, with and without t_{lag}, ten sub-problems representing the different groups, were simulated, each consisting of ten individuals. The concentration profile was simulated based on a sparse design including five sampling times per individual (t = 0.5, 1, 2, 4 and 6 h). The parameters used for the simulations are presented in Table 1. The sampling schedule used was the same used in the valacyclovir studies and the parameter values of Table 1 match roughly the valacyclovir parameters. The compartmental parameters CL, V and k_a, were assumed to be independently and log-normally distributed. An exponential model was used for residual variability at two different levels of 5 % (datasets I and III) and 10 % (dataset II). The non-compartmental parameters were calculated for each individual and the geometric (or arithmetic) means and coefficients of variation per group were derived. The form of the dataset used in this simulation exercise was motivated from a published dataset of valacyclovir in children, see further down for details. All simulations were performed with NONMEM version 7.2 and the non-compartmental parameters were calculated in Matlab version 2013b.

Table 1 Parameter values used for the simulation study

Dataset	Parameter	Mean values	IIV (%cv)
I, II, III	CL (L/h)	20	30
I, II, III	V (L)	60	30
I, II, III	k_a (L/h)	1.5	30
III	t_{lag} (h)	0.3	20
I, III	Residual variability (%cv)	5	–
II	Residual variability (%cv)	10	–

Estimation for simulated data

The two stage approach described above was applied to analyze the data for the various models and scenarios. The mean population compartmental parameters were estimated from the geometric (or arithmetic) means of C_{max}, T_{max}, AUC and $t_{1/2}$ using the FOCE method implemented in NONMEM. Sequentially, the obtained EBEs were used to analyze the means and the coefficients of variation of C_{max}, T_{max}, AUC and $t_{1/2}$ to estimate the inter-individual variability. The performance of the method was evaluated by calculating bias and precision. The percent mean relative bias (RBIAS), was calculated as follows:

$$\%RBIAS = \frac{100}{n}\sum_{i=1}^{n}\frac{\theta_{Ei} - \theta_T}{\theta_T} \tag{7}$$

where θ_{Ei} is the estimated value of the PK parameters for the ith dataset and θ_T is the value of the parameter used in the simulations. Note that parameters θ can be either the means of the PK parameters or the interindividual variabilities. The precision was calculated as the percent relative root mean squared error (RMSE) and as the percent relative absolute error (RAE), respectively as follows:

$$\%RMSE = 100\sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(\frac{\theta_{Ei} - \theta_T}{\theta_T}\right)^2} \tag{8}$$

$$\%RAE = \frac{100}{n}\sum_{i=1}^{n}\frac{|\theta_{Ei} - \theta_T|}{\theta_T} \tag{9}$$

For each of the three datasets (I–III), five estimation methods were used, namely model A using either arithmetic

or geometric means, model B also using either arithmetic or geometric means, and model B ignoring T_{max} value. Also for the scenario with the t_{lag} (dataset III), estimation including and ignoring the t_{lag} was considered. T_{max} was always used in this scenario, so six cases were considered for the t_{lag} dataset. In total 16 cases were evaluated.

The analysis was performed in an automated manner through a Matlab script calling the nmgo.bat command of Wings for NONMEM [8].

Valacyclovir application

Valacyclovir datasets

The simulation study carried out was useful to determine the most appropriate model for the analysis of a real dataset of acyclovir PK after administration of an extemporaneous suspension of valacyclovir to children. The data had similar format to the simulation study and served as a motivating example to the simulation study. Data were retrieved from [6]. Three phase 1, open label, multicentre studies were included, namely studies HS210914, HS210915 and HS210916, having as their primary objective to determine the pharmacokinetics of a valacyclovir oral suspension formulation in children. Participants in each study were stratified according to age: 1–2, 3–5, 6–11, 12–23 months, 2–5 and 6–11 years. Dosing regiments evaluated were 25 mg/kg as a single dose (study HS210914), 10 mg/kg as a twice-daily dose for 3–5 days (study HS210915) and 20 mg/kg as a 3-times-daily dose for 5 days (study HS210916). In total 11 dose or age groups were available. Each dataset came from sampling each child pre-dose and at 0.5, 1, 2, 4 and 6 h after the first dose. Acyclovir concentration–time data were analysed using the standard non-compartmental method and summary statistics were listed by age group (Table 2). As valacyclovir is the prodrug of acyclovir, the equivalent acyclovir dose of the reported valacyclovir dose was used for the analysis, corrected by the respective molecular weights (MW), i.e. valacyclovir MW = 324, acyclovir MW = 225.

Valacyclovir model and priors

The modeling methodology described in Methods section was applied to develop a PopPK model of valacyclovir, i.e. of acyclovir after per os valacyclovir administration. Estimations were performed in NONMEM v 7.2, as described.

Since valacyclovir is a prodrug of acyclovir, what is measured and modelled are acyclovir concentrations, after administration of valacyclovir. Therefore the disposition PK parameters are expected to be similar to the ones of acyclovir with other types of administration, IV or per os acyclovir. In order to estimate the bioavailability of valacyclovir and also obtain a richer covariate model, including renal function and other covariates, informative Bayesian priors were applied on the parameters, taken from the PopPK model of acyclovir found in the paper of Tod et al. [5]. Namely the following covariate model for clearance was used.

$$CL = F_{ac} \cdot \theta_1 \cdot \frac{7.2 \cdot PCA^{\theta_3}}{\theta_2^{\theta_3} + PCA^{\theta_3}} \cdot \frac{BSA}{1.73} \cdot \frac{40}{S_{CR}} \quad (10)$$

where, F_{ac} is the bioavailability of oral acyclovir and was fixed to the reported value in [5], 0.12; PCA is the post conception age in months, BSA is the body surface area in m^2 and S_{CR} is the serum creatinine in μmol/L. Also the following coefficients were taken from [5], 7.2 L/h/1.73 m^2 is the maximum GFR in adults, 40 μmol/L is a typical serum creatinine value and 1.73 m^2 is a typical body surface area value for adults. As age was the only available variable in the dataset, the covariates were expressed in relation to age. In the first stage of the estimation, mean values for the covariates were used, namely, PCA was the mean of each age group plus 9 months, BSA was calculated from the formula $BSA = 0.02465BW^{0.5378}H^{0.3964}$ (Ref. [9]), where body weight, BW, and height, H, took mean values from published age stratified distributions [10] according to the mean age of each group, and S_{CR} was calculated as a function of age from [5] up to 2 years old and from [11] above 2 years old. In Table 3, the 11 age groups are shown with the corresponding values of PCA, BW, H, BSA and S_{CR} used in Eq. 10. Furthermore, informative priors, using the $PRIOR option of NONMEM, were used for θ_1 and θ_3, the sigmoidicity factor, taking the mean values reported in [5], namely 25.5 and 6.17 with the SD of the priors being 2.9 and 0.6 respectively (the standard errors reported in [5]). θ_2, has a meaning of the PCA when half the maximum CL is observed (Table 6).

For the volume of distribution, V, a basic model without covariates was first used, estimating a parameter θ_4 and covariates were considered depending on the trend of the ETAs and based on the OFV drop of more than 6.64 units, which corresponds to a likelihood ratio test with 1 degree of freedom and 1 % significance level.

Bioavaialbility of valacyclovir (F_{val}) was estimated defined by parameter θ_6. IGV was also estimated in order to test for covariates. Estimation of bioavalability from solely per os data became possible by the use of an informative prior on θ_1.

Interindividual variability for CL, V and k_a was estimated in the second stage of the analysis, as described in the Methods section. The covariates for CL (Eq. 10) as well as other important covariates were included in the Monte Carlo simulations through appropriate distributions. PCA was expressed by the uniform distribution in the age range for each

Table 2 Valacyclovir NCA parameter values from [6]

Group	Group age	N	Dose (mg/kg)	AUC (μg*h/mL)			C_{max} (μg/mL)			$t_{1/2}$ (h)			T_{max} (h)	
				GM	Mean	CV	GM	Mean	CV	GM	Mean	CV	Mean	CV
1	1–2 months	9	25	26.9	27.6	0.25	6.88	7.02	0.22	1.99	2.03	0.23	1.87	0.22
2	3–5 months	9	25	16	17.7	0.37	4.67	5.17	0.43	1.69	1.79	0.41	1.58	0.34
3	6–11 months	7	25	13.7	14.1	0.25	4.86	4.94	0.21	1.24	1.25	0.12	1.81	0.27
4	12–23 months	6	10	6.44	6.63	0.27	2.35	2.49	0.37	1.37	1.39	0.22	1.19	0.57
5	12–23 months	6	20	13.29	14.41	0.43	3.82	4.03	0.34	1.97	2.13	0.41	1.25	0.33
6	12–23 months	9	25	14.5	15.3	0.34	4.14	4.67	0.44	1.33	1.33	0.07	1.93	0.48
7	2–5 years	12	10	5.48	5.72	0.3	2	2.05	0.23	1.53	1.65	0.45	1.15	0.47
8	2–5 years	12	20	9.59	10.09	0.33	3.59	3.75	0.3	1.51	1.54	0.23	1.3	0.35
9	2–5 years	12	25	19.8	19.9	0.13	6.4	6.52	0.19	1.44	1.45	0.14	1.21	0.41
10	6–11 years	8	10	8.72	8.84	0.17	3.05	3.08	0.15	1.56	1.58	0.19	1.19	0.45
11	6–11 years	8	20	12.75	13.08	0.26	4.59	4.71	0.25	1.42	1.43	0.11	1.14	0.31

Table 3 Valacyclovir age groups with generated covariates needed for the Eq. 10

Group	Group age	Dose (mg/kg)	PCA (months)	BW (kg)	H (cm)	BSA (m^2)	S_{CR} (μmol/L)
1	1–2 months	25	10.5	4.5	55.3	0.272	41.44
2	3–5 months	25	13	6.5	61.5	0.345	36.19
3	6–11 months	25	18	8.5	69.9	0.420	35.30
4	12–23 months	10	27	11	80.3	0.509	35.01
5	12–23 months	20	27	11	80.3	0.509	35.01
6	12–23 months	25	27	11	80.3	0.509	35.01
7	2–5 years	10	57	15.8	100.75	0.677	40.12
8	2–5 years	20	57	15.8	100.75	0.677	40.12
9	2–5 years	25	57	15.8	100.75	0.677	40.12
10	6–11 years	10	117	29	132.9	1.047	53.59
11	6–11 years	20	117	29	132.9	1.047	53.59

group. For each simulated individual, body weight, *BW*, and height, *H*, were calculated from published age stratified distributions [9] according to the age of the individual, and *BSA* and S_{CR} were derived as previously described.

Model qualification

A non-parametric bootstrap analysis was performed as an internal model validation technique, using the package Wings for NONMEM [8]. A new replication of the original dataset (a bootstrap sample) was obtained by 11 random draws of individual data (with replacement) from the dataset. The final PopPK model was re-fitted to each new dataset and this process was repeated 1000 times with different random draws. The stability of the final model was evaluated by comparing the final model parameter estimates to the median and 95 % confidence intervals of the non-parametric bootstrap replicates of the final model. If the parameter estimates fell within the 95 % confidence interval obtained from the bootstrap analysis the model was considered unbiased. Also the reported standard errors of the model came from this bootstrap analysis. The bootstrap analysis was done only for the first stage of the analysis, the estimation of the means, while for the interindividual variability it was considered too time consuming.

In addition, visual predictive check (VPC) was performed [12] for internal validation of the model. The parameter estimates obtained by fitting the final model were used to simulate 1000 simulated datasets of the same form as the original datasets. A non-parametric 90 % prediction interval around the median of each simulated dataset was constructed to quantify the variability of the model predictions (5th, 50th and 95th percentiles). Further, 95 % confidence intervals were calculated around each of these percentiles from the 1000 replicates. In order to visually compare the model prediction to the observed data the 5th, 50th and 95th percentiles of the data were also calculated, and it was checked whether each of these fall within the 95 % CI of the respective percentiles of the model predictions.

Results and discussion

Simulations

Motivated by the valacyclovir paediatric data (Table 2) and realizing the wider need to produce PopPK models from literature NCA parameter values, derived from sparse PK data such as paediatric data, a simulation study was carried out, to evaluate different methodologies under different scenarios as described in the "Methods" section.

As mentioned in the "Methods" section, 16 simulation runs were carried out among three different scenarios, i.e. with 5 % (dataset I) and 10 % residual error without t_{lag} (dataset II) and with 5 % residual error with t_{lag} (dataset III). The estimations were carried out in two stages. For, the first stage regarding the population means, the results are shown in the upper parts (means) of Tables 4, 5 and 6, for datasets I–III, respectively. In these Tables the %RBIAS, the %RMSE and the %RAE are shown for the three estimated parameters, *CL*, *V* and k_a and especially in Table 6 they are also shown for t_{lag}.

In dataset I, the basic case with 5 % error without a t_{lag} in the simulated data, shown in Table 4 (upper part, means), we can observe the following:

- Overall, in both models A and B, for all parameters, the usage of the geometric means of the NCA parameters seems to be preferable to the arithmetic means when both are available (Table 4, 1st and 3rd columns vs. 2nd and 4th, respectively). For example with model B, the average relative difference (%RBIAS) for CL over the 1000 runs, is −1.8 % when the geometric means are used versus −8.3 % for the arithmetic means, while the corresponding relative absolute differences (%RAE) are 3 versus 8.3 %. In absence of geometric means, the arithmetic means still give useful results.
- When model B is used, the inclusion of T_{max} gives better results than ignoring it (Table 4, 3rd column vs. 5th).
- With the inclusion of T_{max} model B performs slightly better than model A (Table 4, 3rd column vs. 1st), especially in the estimation of volume, *V*, where the difference is pronounced, however model A gives marginally better results for k_a.

In dataset II, when the error of the simulated data is set to 10 %, the following conclusions can be drawn from the results shown in Table 5 (upper part, means):

- All estimates for the parameter means, become less accurate compared to the 5 % error.
- Again when the geometric means are used, the results are less biased compared to the arithmetic means (Table 5, 1st and 3rd columns vs. 2nd and 4th, respectively).
- Also, model B outperforms model A in Clearance, *CL* and Volume, *V*, while the estimates of k_a are significantly biased with both models which seem to have a similar, moderate, performance (Table 5, 3rd column vs. 1st).

In dataset III, with 5 % error but with t_{lag} incorporated in the simulations, we can observe the following from the results shown in Table 6 (upper part, means):

- The geometric means are again preferable and give more accurate estimates compared to the arithmetic means (Table 6, 1st and 4th columns vs. 2nd and 5th, respectively).

Table 4 Dataset I, with 5 % residual error

Means	Model A		Model B		
	GM	AM	GM	AM	GM w/o T_{max}
CL					
%RBIAS	−2.9	−9.3	−1.8	−8.3	−1.8
%RMSE	4.3	9.8	3.6	8.9	3.6
%RAE	3.6	9.3	3	8.3	3
V					
%RBIAS	8.7	11.3	0.3	4.4	4
%RMSE	9.2	11.8	3.3	5.8	6.4
%RAE	8.7	11.3	2.6	4.8	5
k_a					
%RBIAS	3	9.2	−3.3	4.3	7.5
%RMSE	5.4	10.4	6.3	7.1	10.9
%RAE	4.1	9.2	5.2	5.4	8.8
IIV					
CV_{CL}					
%RBIAS	8.9	5.6	−0.6	−8.7	−4.9
%RMSE	15.6	20.6	13	16.2	15.2
%RAE	10.6	10	9.6	13	11.4
CV_V					
%RBIAS	−10	8	−0.3	−3.8	−11.9
%RMSE	15.2	15.5	10.1	11.4	21.1
%RAE	11.5	11.6	7.9	8.8	13.7
CV_{k_a}					
%RBIAS	27.4	37.6	−40	−23.1	29.3
%RMSE	33.7	39.8	42.2	30	55.6
%RAE	26.4	38.6	40	26.4	37.7

The GM columns correspond to the geometric means of NCA parameters being used, the AM when the arithmetic means are used, the "GM without (w/o) T_{max}" corresponds to the case when T_{max} is not included in the data

Table 5 Dataset II, with 10 % residual error

Means	Model A		Model B		
	GM	AM	GM	AM	GM w/o T_{max}
CL					
%RBIAS	−3.9	−10.7	−2.7	−9.8	−2.1
%RMSE	4.9	11.3	4	10.3	3.6
%RAE	4	10.8	3.3	9.8	2.9
V					
%RBIAS	6.8	10	−1.9	3	5.4
%RMSE	7.5	10.7	4	4.7	6.3
%RAE	6.9	10	3.2	3.8	5.5
k_a					
%RBIAS	1.9	9.2	−4.8	3.7	17.5
%RMSE	5.8	11.4	8.2	7.8	18.4
%RAE	4.5	9.3	6.8	6.1	17.5
IIV					
CV_{CL}					
%RBIAS	17.5	15.3	4.3	−4.1	ni
%RMSE	25.4	35.2	17.5	21.4	ni
%RAE	18.5	17.9	11.8	14.9	ni
CV_V					
%RBIAS	−8.9	10.5	1.2	−1	ni
%RMSE	17.2	23.4	13.1	19	ni
%RAE	12.5	17.2	9.3	11.6	ni
CV_{k_a}					
%RBIAS	39.3	53.1	−42.8	−12.4	ni
%RMSE	46.6	58.5	46.6	48.6	ni
%RAE	43.2	54.2	43	34.5	ni

The GM columns correspond to the geometric means of NCA parameters being used, the AM when the arithmetic means are used, the "GM w/o T_{max}" corresponds to the case when T_{max} is not included in the data

ni Not implemented

- Ignoring the t_{lag} when it is present in the data gives more biased results for both models A and B (Table 6, 1st and 4th columns vs. 3rd and 6th, respectively). It is therefore advisable to attempt to estimate the t_{lag}, since the estimates of all parameters become more accurate, even though one additional parameter is being estimated. In a real dataset when it is unknown whether a t_{lag} is present, evidence of its presence can be found in similar datasets from literature or in individual or mean PK profiles that may be available from the specific dataset.
- Model B clearly outperforms model A in this case when t_{lag} is present and is being estimated, in all parameters including t_{lag} itself, for which the model A estimate is very biased while the model B estimate is reasonably accurate (Table 6, 4th column vs. 1st). For example with model B, the average relative difference (%RBIAS) for CL over the 1000 runs, is −0.9 % when the geometric means are used versus −8.1 % with model A, while the corresponding relative absolute differences (%RAE) are 2.4 versus 8.1 %. Also, for t_{lag}, the average relative difference (%RBIAS) with model B, is −0.2 versus −129.9 % with model A, while the corresponding relative absolute differences (%RAE) are 10.3 versus 129.9 %.

Moving on to the second stage, the estimation of the interindividual variability of the compartmental PK parameters, using both the means and the CVs of the NCA parameters, the results are shown in the lower parts (IIV) of Tables 4, 5, 6 for datasets I–III, respectively. For dataset I (Table 4) the following conclusions can be summarized:

- The geometric means are again preferable to the arithmetic means and give more accurate estimates in most cases. The exception is the CV for the parameter k_a, using model B.
- Model B performs better for *CL* and *V* than Model A, but not for k_a using the geometric means. When the arithmetic means are used with model B, k_a is estimated with similar accuracy to model A using the geometric means. However it could be argued that the accuracy of estimating the interindividual variability of k_a is not important enough, to change the general conclusion from the previous results that model B with the geometric means is the choice that performs best. The interindividual variability of the absorption rate constant, k_a, is a parameter difficult to estimate in PopPK analysis, even when dense raw data are available.

For dataset II when the error is increased to 10 %, the results shown in Table 5 (lower part, IIV) are similar to the ones from dataset I with 5 % error, but with slightly increased biases. However the general picture is that reasonable estimates of the IIV of CL and V can be obtained, but significant biases are present in the estimates of IIV of k_a.

Finally for dataset III, when the error is 5 % but t_{lag} is incorporated into the simulated dataset, the general picture does not change dramatically (Table 6, lower part, IIV).

- Usable estimates seem to be obtainable for the IIV of *CL* and *V* and less so for k_a. There was no attempt to estimate the IIV of the t_{lag} parameter.
- Despite the fact that the CV of k_a is not estimated very accurately, the results indicate (Table 6, 7th column) that omitting the estimation of that parameter, has a negative influence in the estimates of the interindividual variabilities of clearance and volume, which are usually more important, therefore it is better to include it in the estimation.

Table 6 Dataset III, with 5 % residual error and lag time incorporated in the simulations

Means	Model A			Model B			
	GM	AM	GM w/o t_{lag}	GM	AM	GM w/o t_{lag}	GM w/o CV_{k_a}
CL							
%RBIAS	−8.1	−3.1	−8.9	−0.9	−8.2	−1.3	–
%RMSE	8.6	4.4	9.4	3	8.7	3	–
%RAE	8.1	3.5	8.9	2.4	8.2	2.4	–
V							
%RBIAS	20.6	14.4	4.2	−3.1	10.8	−22.2	–
%RMSE	21.1	14.9	5.5	4.8	12.2	23.3	–
%RAE	20.6	14.4	4.6	3.8	10.8	22.2	–
k_a							
%RBIAS	69.7	23.1	−27.6	−13.8	35.8	−47.3	–
%RMSE	71.7	23.3	27.7	16.1	43.8	47.7	–
%RAE	69.7	23.1	27.6	14.2	35.8	47.3	–
t_{lag}							
%RBIAS	129.9	65.7	na	−0.2	86.3	na	–
%RMSE	132.1	67.3	na	13.4	91.8	na	–
%RAE	129.9	65.7	na	10.3	86.3	na	–
IIV							
CV_{CL}							
%RBIAS	−52.5	32.4	ni	−1.8	−11.7	ni	−6.7
%RMSE	57.6	49.2	ni	15.9	21.2	ni	24.7
%RAE	53.2	32.5	ni	11.6	17.3	ni	15.2
CV_V							
%RBIAS	−63.9	−15	ni	−3.8	−20.4	ni	33
%RMSE	66.8	22.6	ni	23.1	27.4	ni	38.5
%RAE	63.9	16.3	ni	16.1	23.4	ni	35.8
CV_{k_a}							
%RBIAS	−59.1	80.7	ni	45.3	56.5	ni	na
%RMSE	63.8	83.9	ni	49.2	60.7	ni	na
%RAE	61.2	80.8	ni	45.3	56.9	ni	na

The GM columns correspond to the geometric means of NCA parameters being used, the AM when the arithmetic means are used, the "GM w/o t_{lag}" corresponds to the case when t_{lag} is not included in the model, the "GM w/o CV_{k_a}" column corresponds to the case when IIV only for *CL* and *V* is being estimated

na Not applicable

ni Not implemented

In Fig. 1, a visual predictive check plot of model B with t_{lag} fitted to the geometric means of the NCA parameters of a representative single study taken from dataset III is shown. Ninety percentage prediction intervals (5th, 50th, 95th percentiles from 1000 virtual patients) overlying together with the PK profiles of the dataset (100 individuals) demonstrate that the model describes well the raw data despite the fact that the parameters have been estimated from NCA summaries of these data (See Fig. 1 legend for parameter estimates). The slight underprediction of interindividual variability can be attributed to the fact that no residual error has been added to the prediction intervals of the model since this is not available, while the simulated dataset did carry a 5 % residual error. Also, note that the lines simply connect the data and do not shape a profile, this is why the t_{lag} is not visible.

Overall, the conclusions from the simulation study are that model B, which takes explicitly into account the sampling schedule, fitted to data of NCA parameters expressed as geometric means, is the best choice to convert NCA parameters to one-compartment model parameters. The advantage of model B is possibly diminished when the NCA parameters come from dense data, but seems to be significant for sparse data such as the ones from paediatric PK studies. Clearly, unlike the simulated data used here,

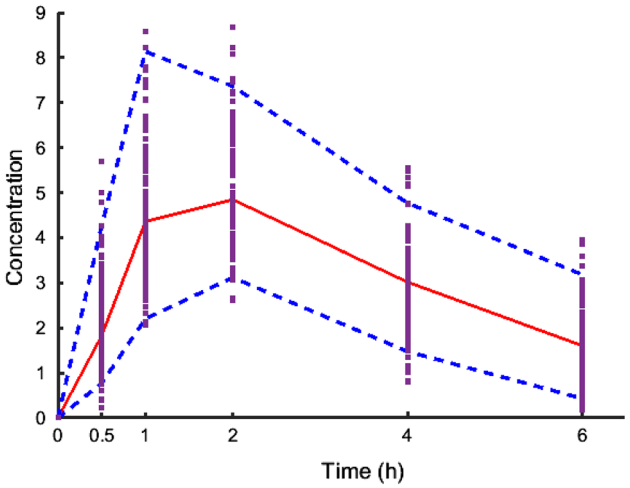


Fig. 1 Visual predictive check plot of model B with t_{lag} fitted to the geometric means of the NCA parameters of a single representative simulated study from dataset III. 90 % prediction intervals (5th, 50th, 95th percentiles, 1000 virtual individuals) overlying together with the PK profiles of the dataset (100 individuals) are shown. The following model parameters estimates were used: CL = 20 L/h, V = 59.1 L, $k_a = 1.38\ h^{-1}$, $t_{lag} = 0.315$ h, $CV_{CL} = 34.5$ %, $CV_V = 30.7$ %, $CV_{ka} = 46.1$ %

real pharmacokinetic data, do not follow ideal one compartment kinetics, so there will be an added bias from model misspecification. These techniques are intended for drugs that their PK has been studied adequately rather than for novel compounds. The PK model should be known a priori to be one-compartment and as mentioned, the presence or not of a t_{lag} can also be speculated from available individual profiles. It seems to be difficult to carry out model selection solely based on the NCA data. Although this approach could be expanded for the two-compartment model, this would demand additional NCA information such as AUMC which is not frequently available in literature, so its applicability would be limited.

Valacyclovir

This work was motivated by the need to develop a preliminary PopPK model for valacyclovir in children which could be used to carry out simulations in order to predict drug profiles in children and design new prospective clinical studies. The model chosen to analyze the NCA parameter values according to the results of the simulation study was model B, while evidence that a t_{lag} is present was drawn from individual PK profiles from the same [13] and other [14] clinical studies, that were available in literature.

Initially, model B was fitted using the covariate model of Eq. 10 with informative priors on parameters θ_1 and θ_3. The prior on $\theta_1$was included in order to estimate the bioavaiability θ_6 and the prior on θ_3 was included since the dataset used cannot support on its own such a complex covariate model. From the first stage of the analysis, as described in the methods section, by fitting the model to the geometric means of C_{max}, AUC, T_{max} and $t_{1/2}$ shown in Table 2, parameters θ_1 to θ_7 were estimated together with intergroup variability for V and F_{val}, among the 11 groups. The parameter estimates of the basic model are shown in Table 7 together with the corresponding relative standard errors (RSE).

The main available independent variable from the data was PCA. From PCA, H, BW and BSA, as well as S_{CR} were simulated as described in the "Methods" section and shown in Table 3.

For volume, BW was found to be a significant covariate explaining intergroup variability. The covariate was modeled with a power law, centered on the BW value used in [5], namely

$$V = F_{ac}\theta_4\left(\frac{BW}{6.9}\right)^{\theta_8} \quad (11)$$

OFV dropped from 25.616 (basic model) to 8.908 (final model). Intergroup variability became close to zero and was dropped from the final model, while its main purpose was the detection of potential covariates. Also for bioavailability, no significant covariate was detected. This is despite evidence that the enzymes responsible for the hydrolysis of valacyclovir to acyclovir as well as the transporters responsible for the absorption of valacyclovir in the gut may be immature in younger ages such as in infants. The final model parameter estimates are shown in Table 7 together with the corresponding RSEs.

In the second stage of the analysis, the IIV estimates expressed as CVs were derived by fitting the final model structure as described in the methods section to the CVs of the 4 NCA parameters together with their geometric means. The detected covariates for intergroup variability were considered to contribute to interindividual variability within the age groups. These covariates were taken into account in the estimation of the CVs of PK parameters, as described in the methods. The results of the second stage are reported in Table 7 with their RSEs.

In Table 7 the results of the bootstrap analysis from 1000 bootstrap samples as described in the methods section are included, for the first stage of the analysis and for the final model. Bootstrap RSEs are shown as well as medians and 95 % CI. The NONMEM estimates lie within the 95 % CI.

Observed versus predicted plots of the NCA parameter means and their CVs are shown in Fig. 2. Figure 2A is on a log–log scale in order to better depict the DVs of different orders of magnitude. Note that there are four different kinds of DVs (C_{max}, AUC, T_{max} and $t_{1/2}$) with different units and different orders of values. Also the exponential

Table 7 Parameter estimates for the valacyclovir PopPK model

Parameter	Prior (SD)	Basic model		Final model		Bootstrap		
		Estimate	% RSE	Estimate	% RSE	Median	% RSE	95 % CI
Stage 1: estimation of the means								
θ_1 (L/h)	25.5 (2.9)	23.1	4.2	24.6	1.1	24.6	1.4	(23.7, 25.1)
θ_2 (Months)	–	10.2	9.1	10.4	2.9	10.5	17.5	(9.2, 14.1)
θ_3	6.17 (0.6)	6.17	0.38	6.12	1.0	6.13	1.6	(5.8, 6.2)
θ_4 (L)	–	102	14.3	71.7	10.1	73.5	14.2	(59.0, 91.4)
θ_5 (k_a, h^{-1})	–	5.18	37.1	3.8	28.2	3.63	33.2	(1.7, 6.3)
θ_6 (F_{val})	–	0.488	8.4	0.53	7.4	0.52	7.4	(0.4, 0.6)
θ_7 (t_{lag}, h)	–	0.914	5.9	0.68	14.5	0.82	18.6	(0.4, 0.9)
θ_8	–	–		0.79	16.9	0.8	19.9	(0.5, 1.1)
IGV_V (%)	–	39.4	43.4	–	–	–	–	–
$IGV_{F_{val}}$ (%)	–	17.6	49.8	–	–	–	–	–
σ (%)	–	15.2	11.6	20.7	10.3	18	13.6	(13, 23)
OFV	–	25.616	–	8.908	–	–	–	–
Stage 2: estimation of the IIV								
CV_{CL} (%)	–	–	–	21.8	12	–	–	–
CV_V (%)	–	–	–	22.3	12.7	–	–	–
CV_{k_a} (%)	–	–	–	55.6	10.4	–	–	–
σ_1 (%)	–	–	–	20.3	9.9	–	–	–
σ_2 (%)	–	–	–	42.6	9.3	–	–	–

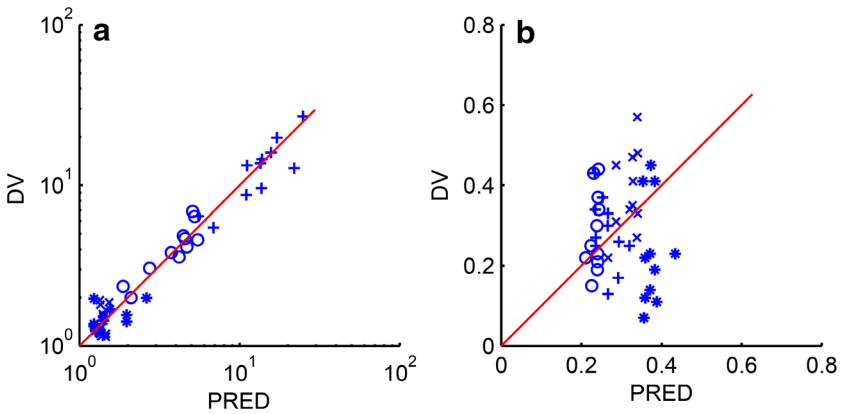


Fig. 2 Observed versus predicted of geometric means (**a**), and CVs (**b**) of the NCA parameters of valacyclovir. Key: C_{max} (*circle*), *AUC* (*plus symbol*), $t_{1/2}$ (*asterisk*) and T_{max} (*times*)

error is used to account for this heterogeneity in the DVs. On the other hand all values shown in Fig. 2b are CVs and fall within a narrow range of values. Dispersion of the DV values is wide but roughly symmetrical along the unit line. Further, in Fig. 3, VPC plots for all parameters are shown. The median and 90 % prediction intervals (the 5th and 95th percentiles of the predictions) together with 95 % CI around each of the percentiles, as well as the median, the 5th and 95th percentiles of the data, are shown. It can be observed that for all parameters except the half-life, the data lie within the 95 % CI of the predictions. For the half-life the median of the means is within CI but the 5th and 95th percentiles are marginally outside (plot 3C). Also for the CV of the half-life all predicted percentiles are over-predicted compared to the data (plot 3G).

In Fig. 4 values of clearance of acyclovir after IV administration, with respect to age, taken from [7], are shown, co-plotted with the 90 % prediction intervals of simulated clearance values from the model. This graph shows that clearance per body weight is rising up to a maximum reached at age 1 year, while for older children the clearance per body weight drops, as expected, down to the final adult value. This trend is captured by the model. The IV acyclovir data shown in Fig. 4 have not been used in the development of the model,

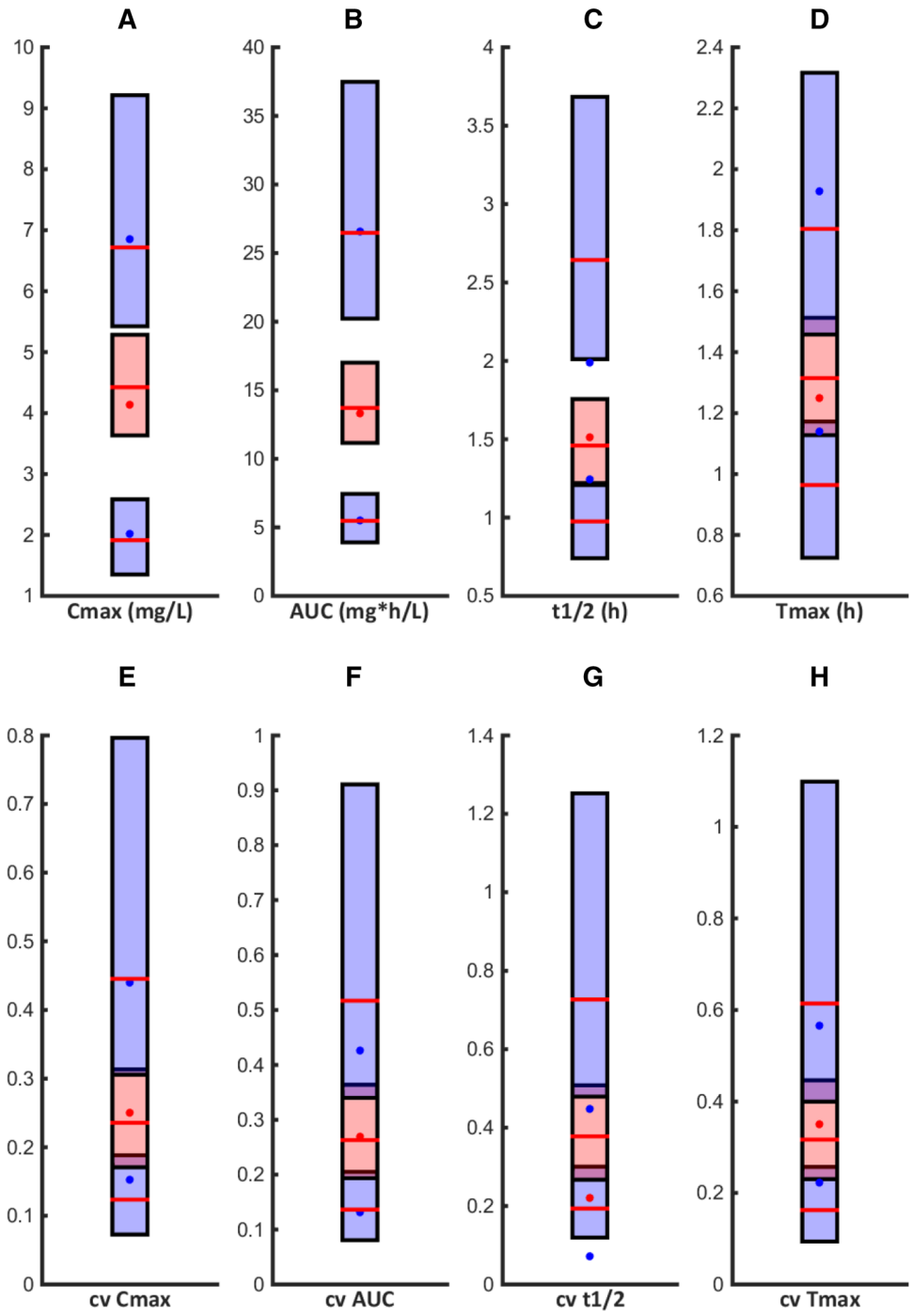


Fig. 3 Visual predictive check plots for the geometric means (A: C_{max}, B: AUC, C: $t_{1/2}$, D: T_{max}) and CVs (E: CV-C_{max}, F: CV-AUC, G: CV-$t_{1/2}$, H: CV-T_{max}) of the NCA parameters of valacyclovir. *Dots* are the 5th, 50th and 95th percentiles of the data. *Boxes* represent the 95 % CI for the 5th, 50th and 95th percentiles of the prediction interval, where *red lines* are the medians (Color figure online)

and as such, they may be regarded as an external validation of the clearance model derived here.

Comparing the valacyclovir model with the acyclovir model of [5], one may notice the significant difference in the value of volume. However the value of volume estimated here is in accordance to the half-life of the drug as derived by the formula $t_{1/2} = V \log 2/CL$. For example, for weight 10 kg, from Eq. 11 the volume works out to be 11.5 L while in [5], it is 6.4 L. However, in [5], it seems that the half-life of the drug of around 2.5 h, is not determined by the ratio volume over clearance but instead from 1/Ka, being in a "flip-flop" regime.

Overall the model seems to pass the usual internal validation tests and is considered to describe the data adequately. The PopPK model for valacyclovir developed here, is a preliminary model that was put together by integrating literature information that was available in various forms, and it is not a classic PopPK since it has not

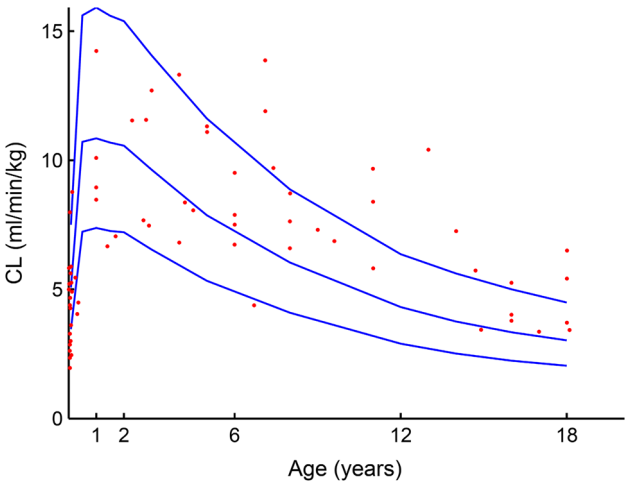


Fig. 4 Clearance values of acyclovir after IV administration (data not included in the development of the model) with respect to age from 0 to 18 years, taken from [7], co-plotted with the 90 % prediction interval of simulated clearance values from the model

been developed from raw PK data. The model has been used to carry out simulations to provide dosage recommendations for children of various ages, by considering therapeutic targets that were calculated by the prescribed acyclovir doses. The extrapolation was carried out as follows, but the results are not shown here and are published separately: Since acyclovir is prescribed in children, assuming that the recommended doses of acyclovir are efficacious, a PopPK model of acyclovir [5] was used to carry out simulations to determine a critical concentration, C_{crit}, for which a target criterion is fulfilled, i.e. 90 % of the simulated patients have acyclovir levels above C_{crit} for at least 50 % of the time. The same was done for a secondary target, drug exposure, determining AUC_{cirt}. Then, the present PopPK model of valacyclovir was used to determine by simulations, dosage regimens that fulfil the criteria for both targets. This was repeated for various indications and age groups, including renally impaired patients.

Further, the model can be used to carry out simulations in order to design efficiently prospective clinical trials, that will produce PK data to build a proper PopPK model for valacyclovir in children.

Conclusion

Motivated by the need to develop a usable PopPK model of valcyclovir in children from NCA parameter values found in literature, a simulation study was carried out to evaluate the performance of the various modelling choices that had to be made to develop the model for valacyclovir. The conclusions from the simulation study are that a model which takes explicitly into account the sampling schedule, fitted to geometric means of NCA parameter values, performs best for the sparse paediatric sampling schedule considered. The mixed effects, meta-analysis approach utilized has several advantages apart from accounting properly the random intergroup variability and the detection of covariates. It also gives the potential to apply informative Bayesian priors on the parameters in order to integrate to the model, additional information available in the form of parameter values. In the example of the valacyclovir model developed, informative priors from a PopPK model of per os acyclovir were applied for some of the parameters related to acyclovir clearance, in order to include a richer covariate model for clearance, not supported by the NCA dataset and a value for bioavailability. The application to valacyclovir presented in this work, provides a preliminary model that aims to demonstrate the usage of the methodology. The model itself can, and has been used in simulations to provide dosage recommendations for children of various ages and to help design more efficiently prospective clinical trials.

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